Natural Language Processing
(Sem-VII/ BTech-IT)

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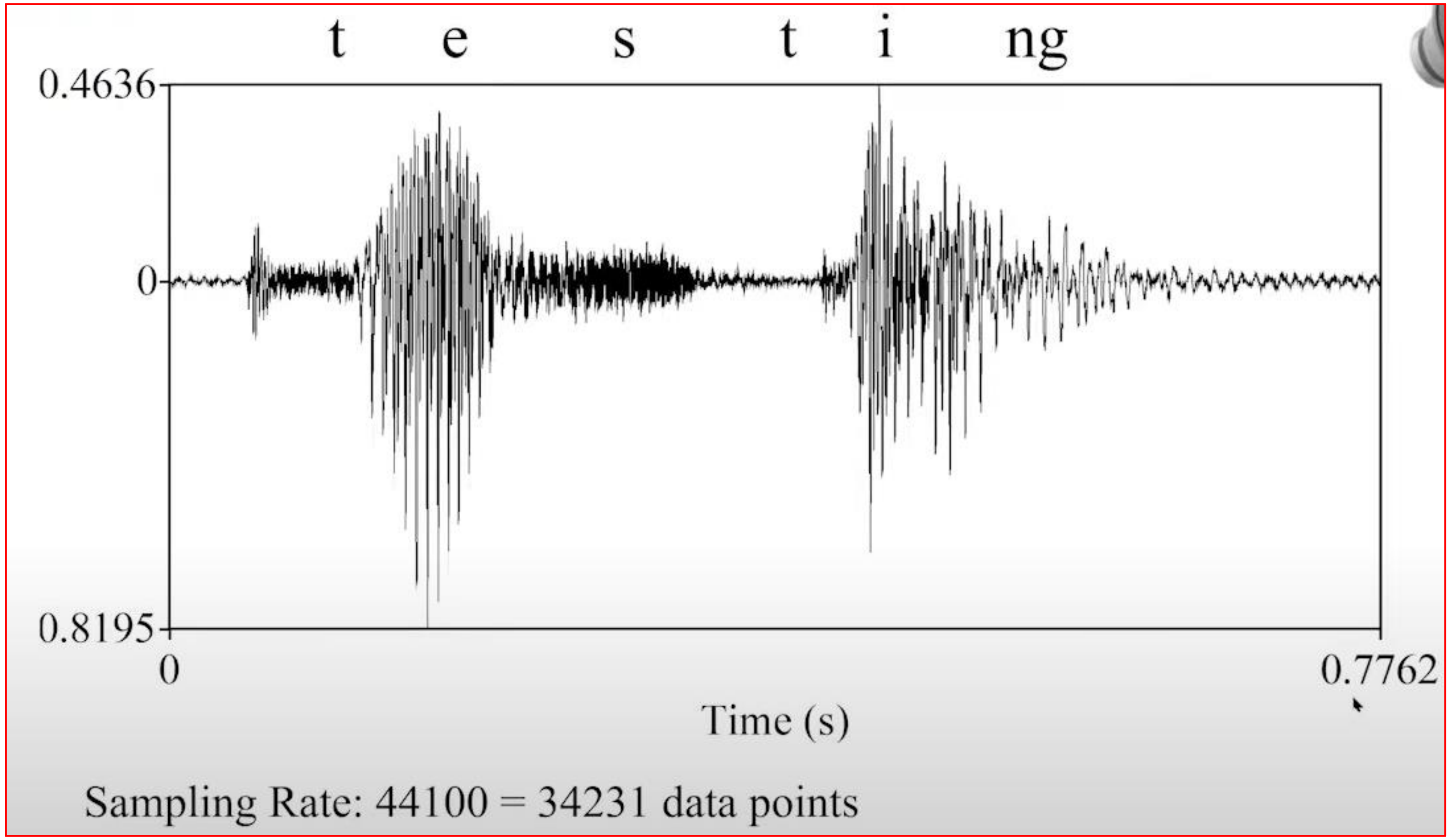
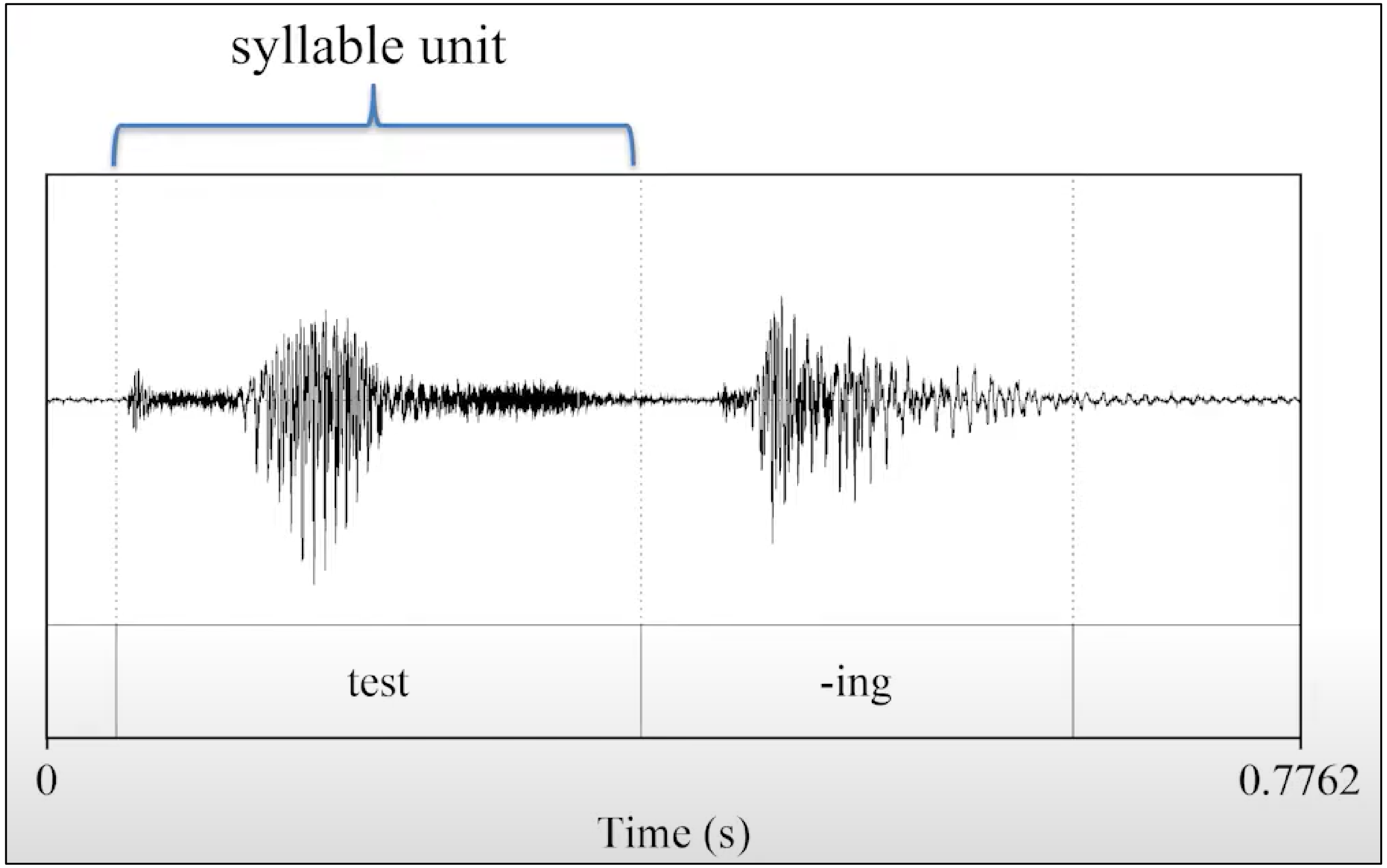
Unit-1-Introduction

Module No.	Unit No.	Details	Hrs	CO
1	**Basics of NLP**		10	CO1
	1.1	Biology of Speech Processing; Place and Manner of Articulation		
	1.2	Word Boundary Detection		
	1.3	Argmax based computations		
	1.4	HMM and Speech Recognition		

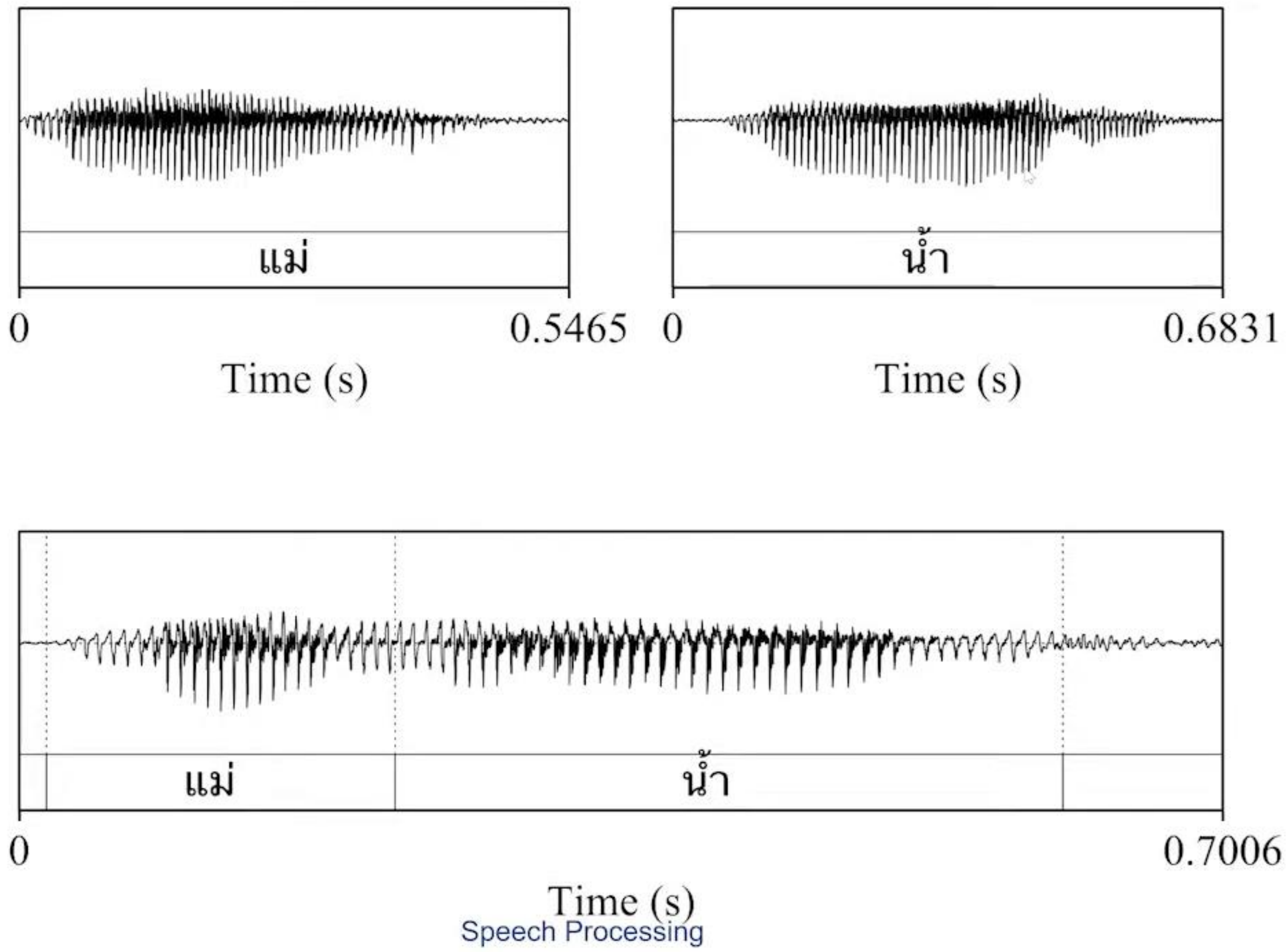
Speech Signal is Time Domain



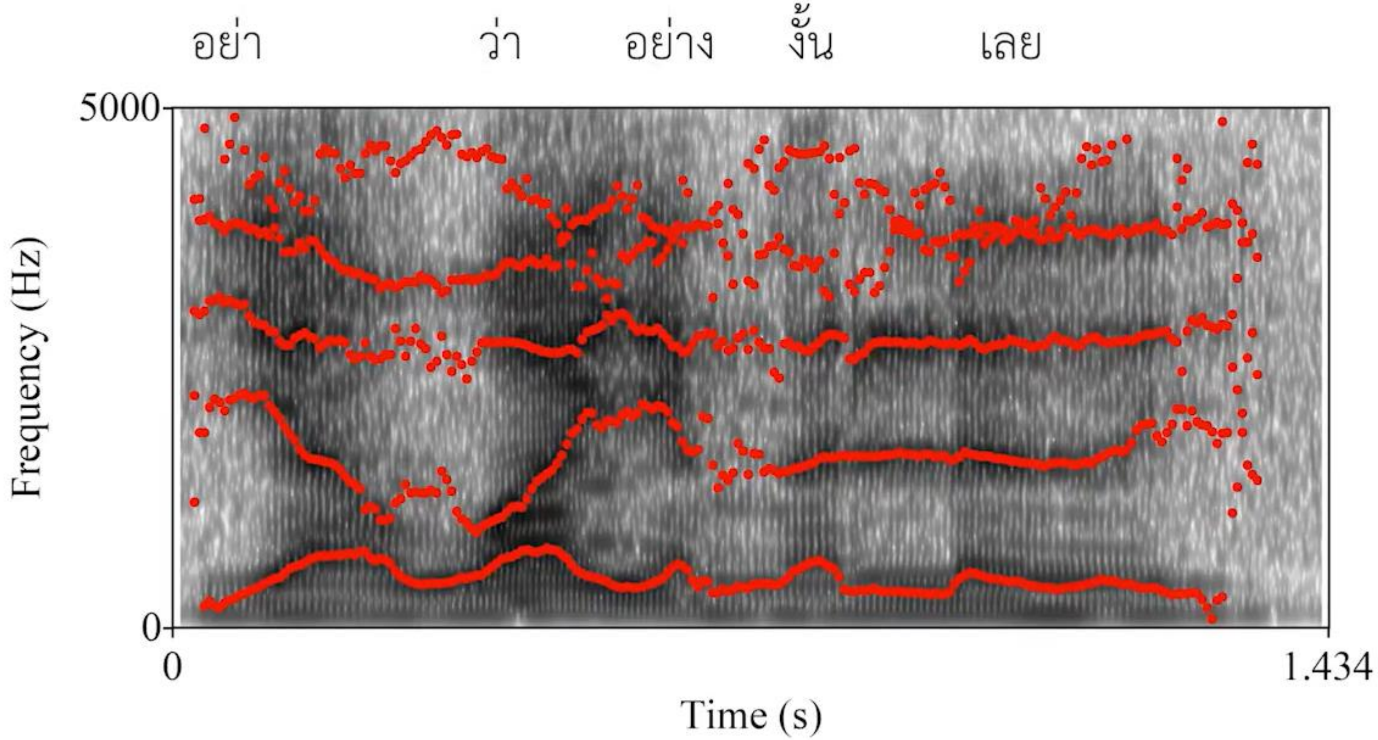
Syllables



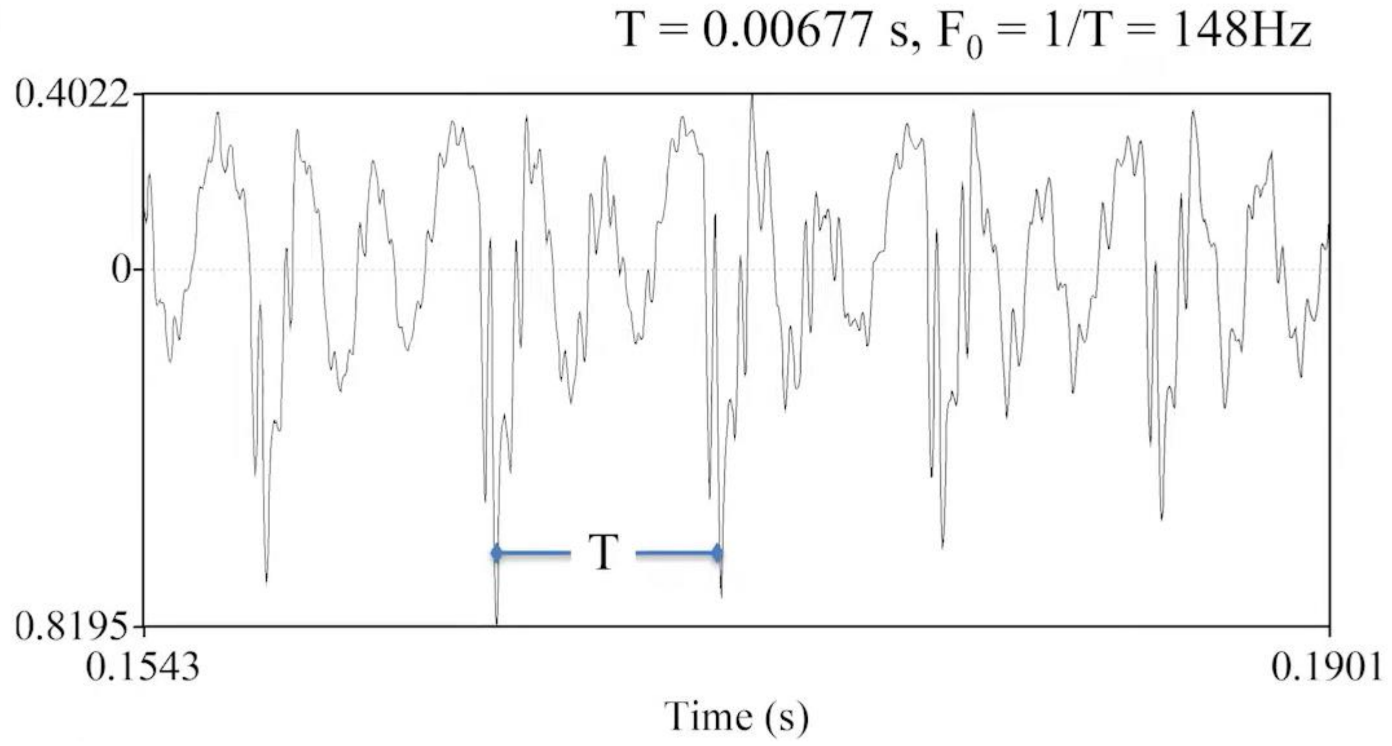
Word Stress Duration



Spectral Continuity

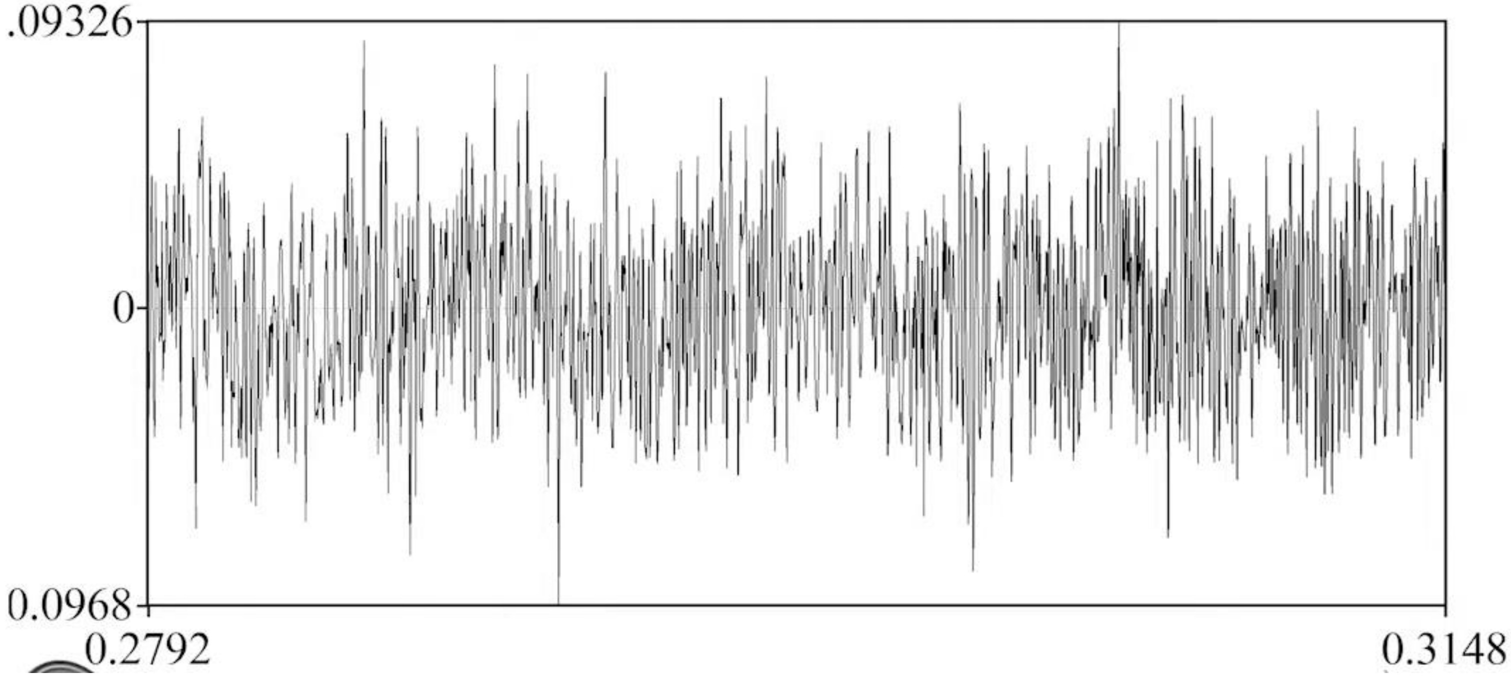


“Testing” - / e / sound

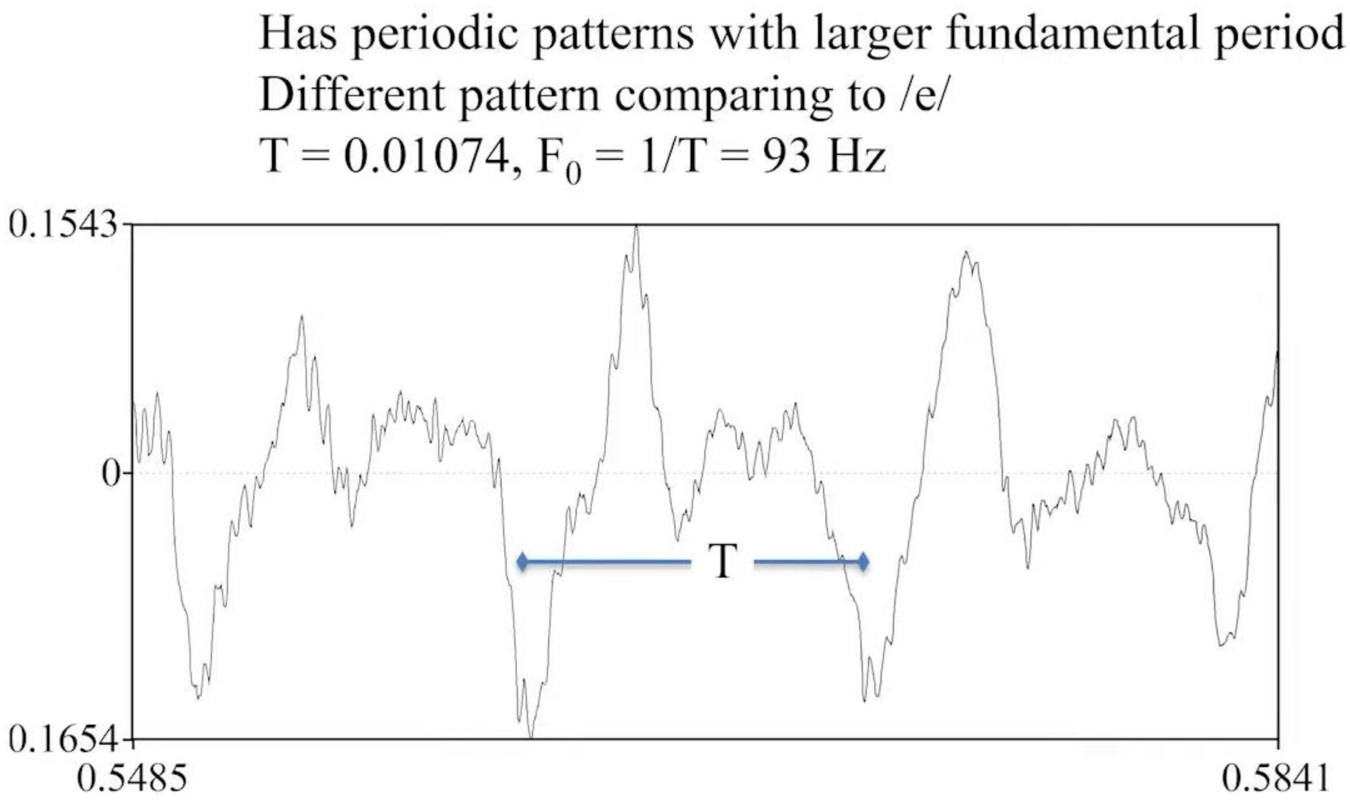


“Testing” - / s / sound

No obvious periodic patterns

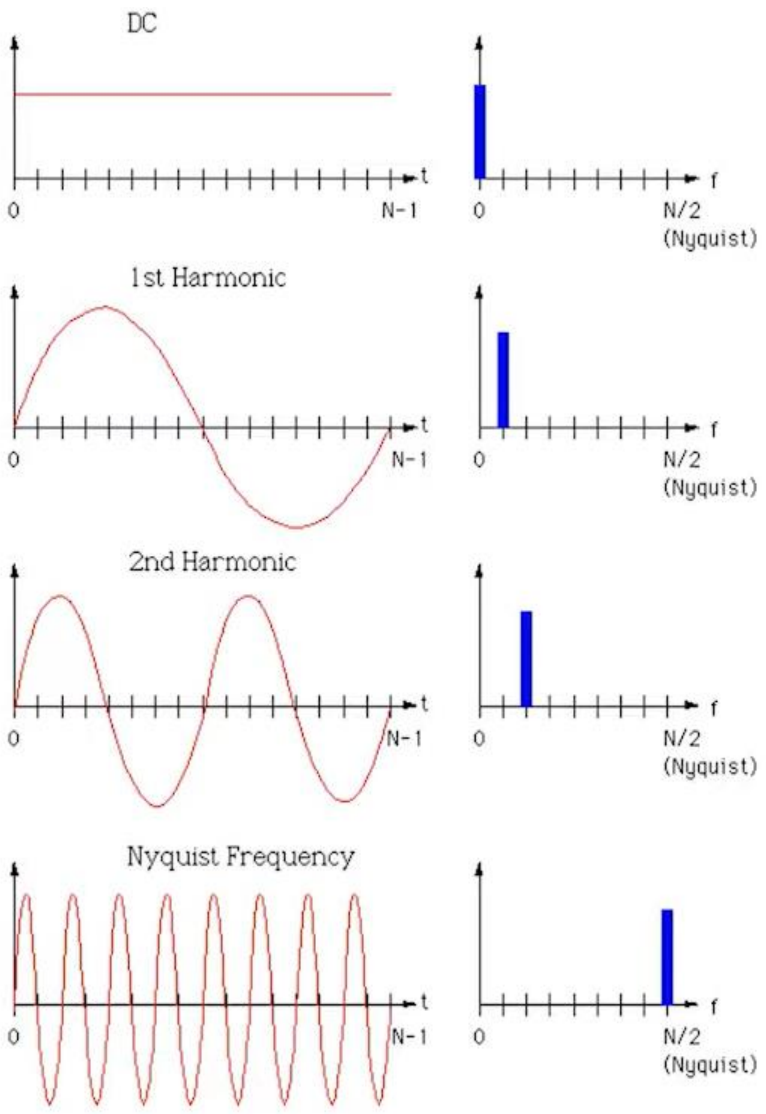


“Testing” - / i / sound



Spectrum

- If we think of a signal to compose of multiple sinusoidal components, different time structures can be thought of as having different amount of frequency components
- To analyze frequency component of signals, we use Fourier transform technique
- Discrete Fourier Transform (DFT) is the DSP technique for analyzing the frequency domain components (a.k.a. spectrum)



Discrete Fourier Transform

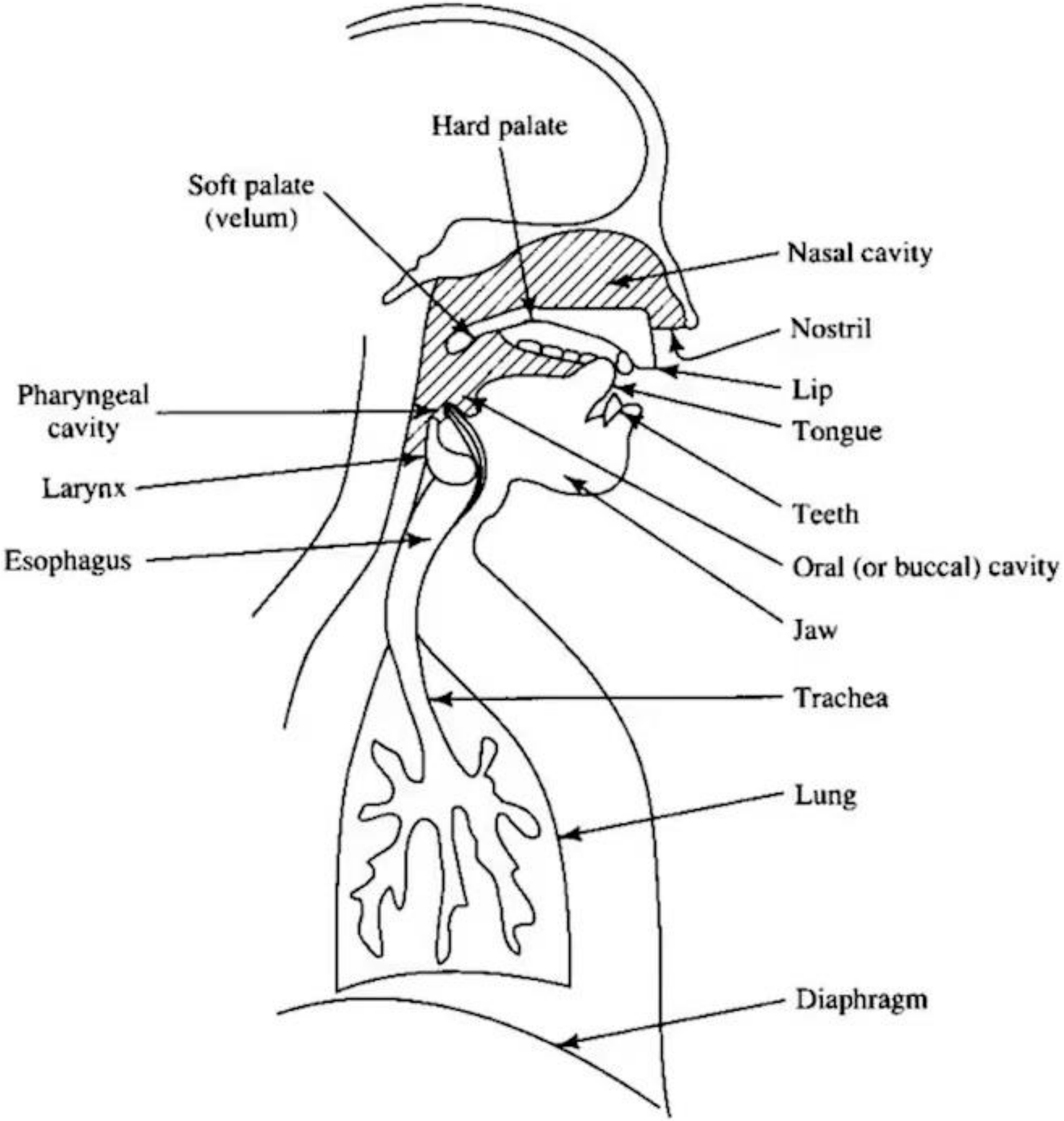
$$X[k]=\sum_{n=0}^{N-1}x[n]w^{kn},\ \ w=e^{-j2\pi/N}$$

Time domain signal

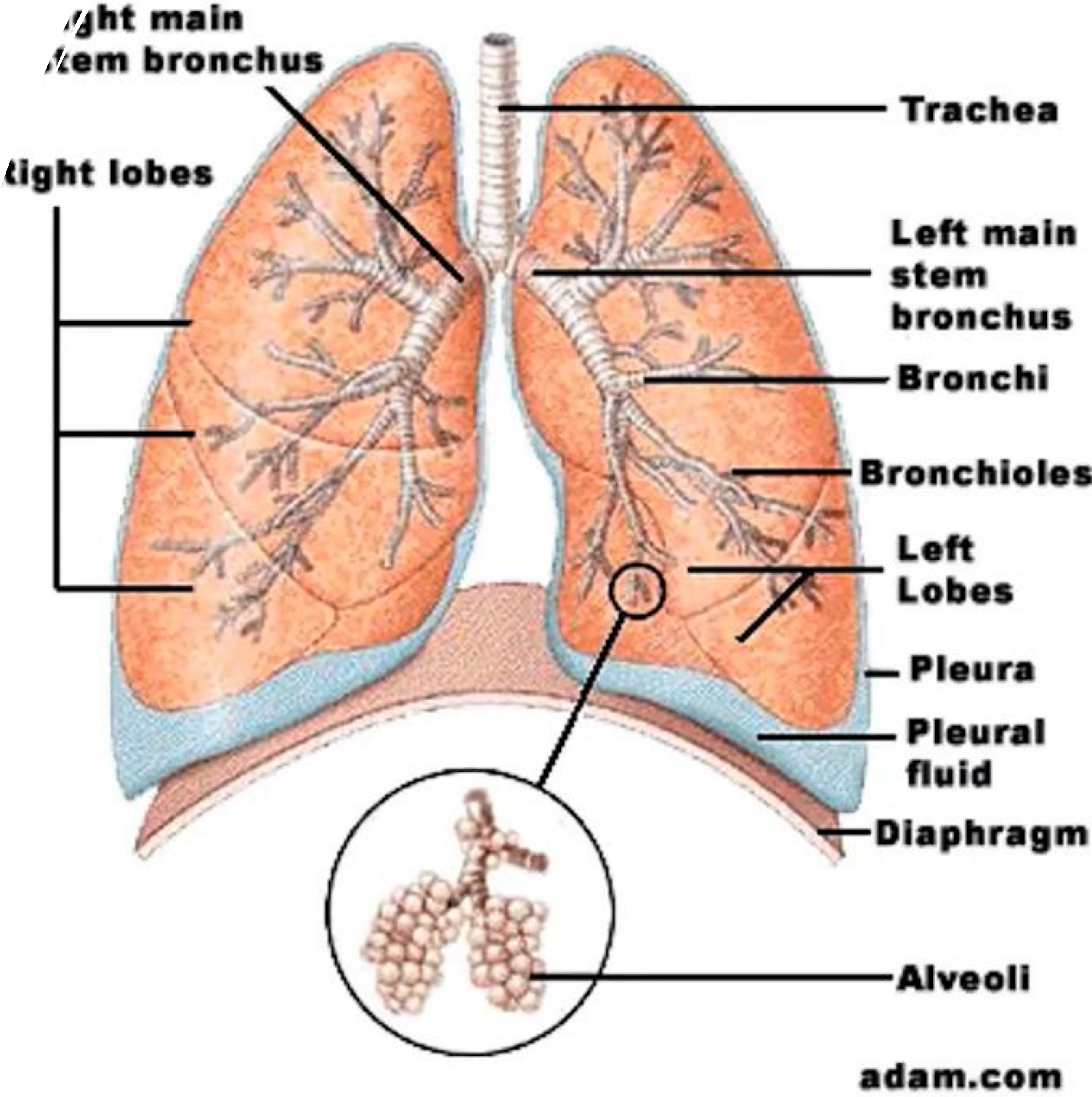
Spectrum at frequency

$$\frac{k}{N/2}\cdot\frac{F_s}{2}=\frac{kF_s}{N}\text{Hz}$$

Anatomy of Human Speech Production System

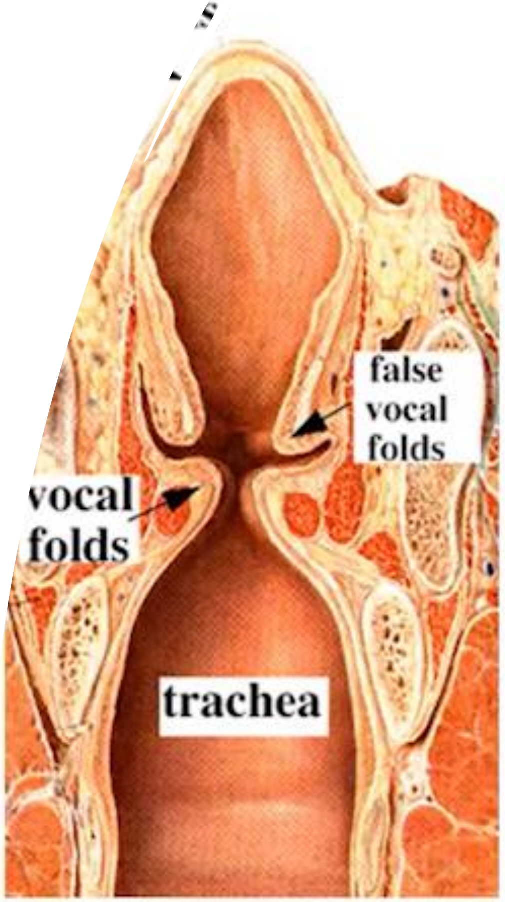


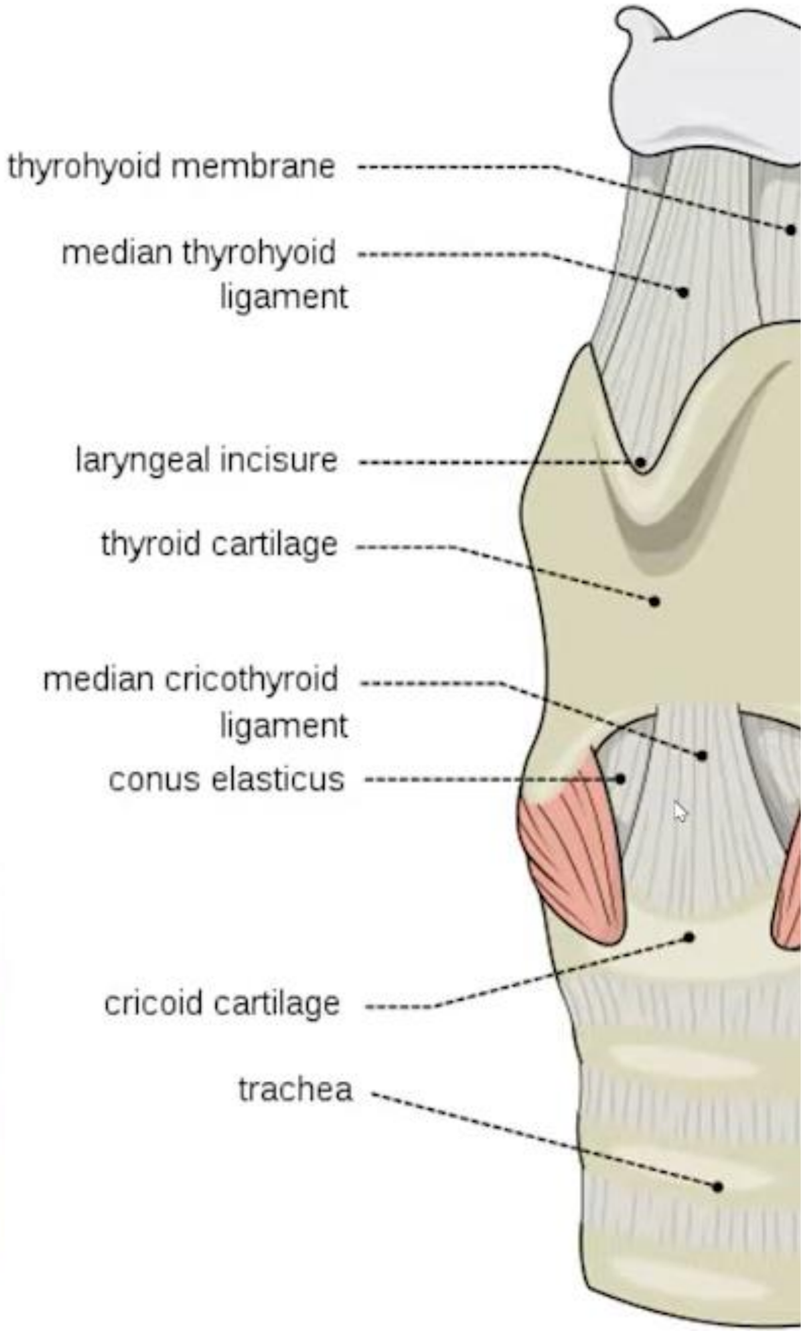
Lung



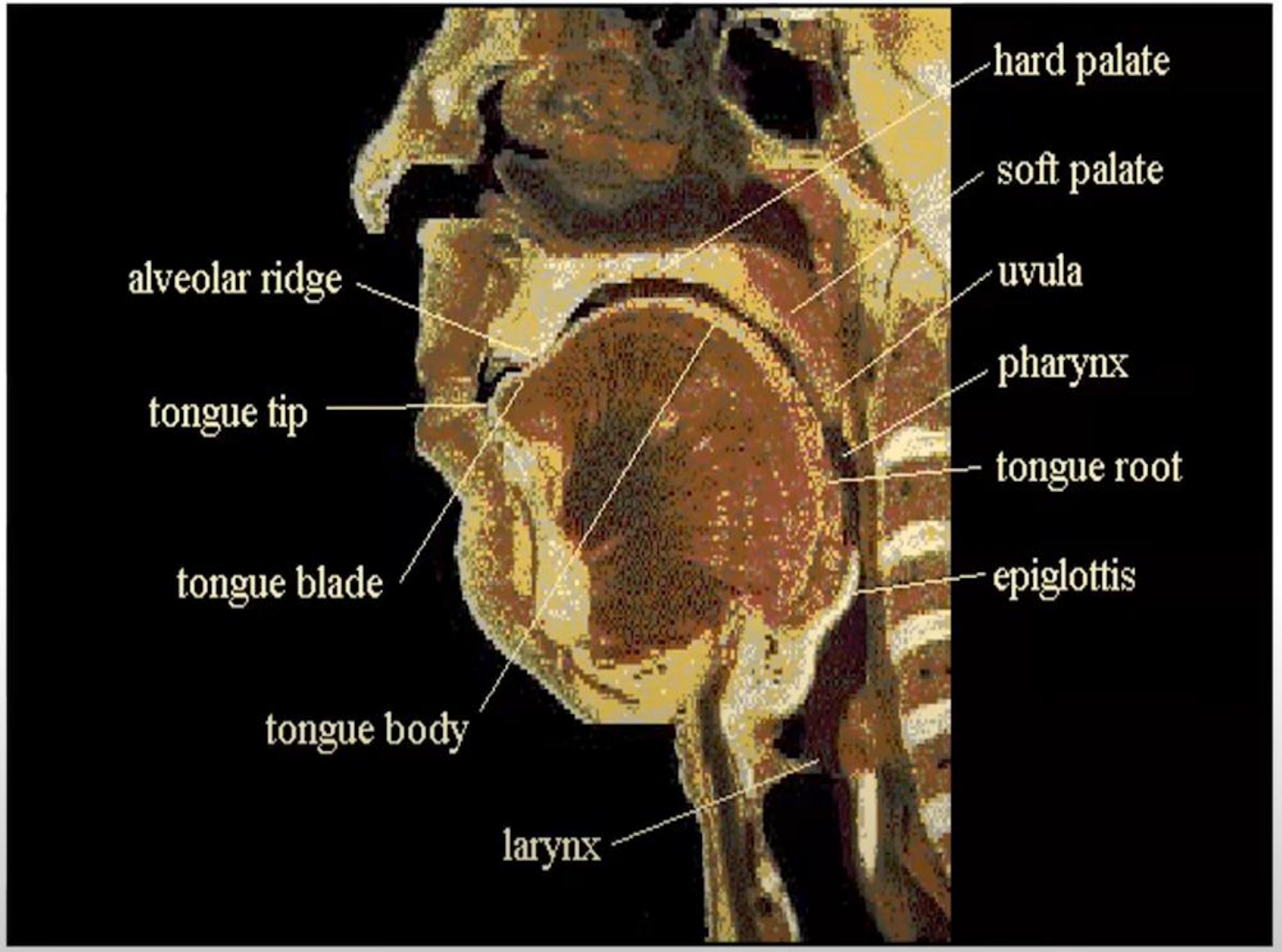
ght main
tem bronchus
Trachea
ight lobes
Left main
stem
bronchus
Bronchi
Bronchioles
Left
Lobes
Pleura
Pleural
fluid
Diaphragm
Alveoli
adam.com

Larynx


false vocal folds
vocal folds
trachea


thyrohyoid membrane
median thyrohyoid ligament
laryngeal incisure
thyroid cartilage
median cricothyroid ligament
conus elasticus
cricoid cartilage
trachea

Larynx



hard palate
soft palate
uvula
pharynx
tongue root
epiglottis
alveolar ridge
tongue tip
tongue blade
tongue body
larynx

Place and Manner of Articulation in NLP

Basic Definition

- **Phonetics**: The study of human speech sounds.
- **Articulation**: How speech sounds are formed using parts of the mouth and vocal tract.
- Two key dimensions:
  - ✓**Place of Articulation** – *where* in the mouth the sound is produced.
  - ✓**Manner of Articulation** – *how* the sound is produced.

Relevance in NLP

Understanding articulation helps machines:

- ✓Recognize spoken words more accurately.
- ✓Generate more natural-sounding speech.
- ✓Handle accents, pronunciation variations, and mispronunciations.

What is Place of Articulation?

- “Place of articulation” refers to the **physical location** in the mouth or vocal tract where a sound is produced.
- **/p/ vs /f/** — “pat” vs “fat” — lips vs lip + teeth.

Types of Place of Articulation

Place of Articulation	Description	Example Sounds	Example Words
Bilabial	Both lips touch	/p/, /b/, /m/	"pat", "bat", "mat"
Labiodental	Bottom lip + upper teeth	/f/, /v/	"fan", "van"
Dental	Tongue against teeth	/θ/, /ð/	"think", "this"
Alveolar	Tongue at alveolar ridge	/t/, /d/, /n/, /s/, /z/	"top", "dot", "no", "sip", "zip"
Postalveolar	Tongue just behind ridge	/ʃ/, /ʒ/	"ship", "measure"
Palatal	Tongue to hard palate	/j/	"yes"
Velar	Tongue to soft palate	/k/, /g/, /ŋ/	"cat", "go", "sing"
Glottal	At the vocal cords	/h/, glottal stop	"hat", (sound between "uh-oh")

What is Manner of Articulation?

- Refers to how the airstream is modified when producing a sound.
- Defines how the vocal tract constricts airflow.
- **Contrast Example:**
  - ✓ **Stop vs Fricative**: “tip” (/t/) vs “sip” (/s/)
  - ✓ **Affricate**: "chocolate" begins with /tʃ/, a combination of stop and fricative.

Types of Manner of Articulation

Manner	Description	Example Sounds	Example Words
Stops (Plosives)	Complete stop and release of airflow	/p/, /t/, /k/, /b/, /d/, /g/	"pen", "top", "go"
Fricatives	Narrow airflow creates friction	/f/, /v/, /s/, /z/, /ʃ/, /ʒ/, /h/	"fish", "zoo", "she"
Affricates	Stop + Fricative combo	/tʃ/, /dʒ/	"chip", "judge"
Nasals	Air flows through the nose	/m/, /n/, /ŋ/	"man", "no", "song"
Liquids	Smooth airflow around tongue	/l/, /r/	"lip", "red"
Glides	Slight movement like vowels	/w/, /j/	"win", "yes"

Real-World Applications in NLP

Speech Recognition:

- Systems need to distinguish between similar-sounding words like “bat” and “pat”.
- Understanding articulation helps them recognize phonemes accurately.

Text-to-Speech (TTS):

- To generate natural-sounding voices, the system must simulate place and manner of sound production.
- Example: Google Assistant generating clearer /s/ or /t/ sounds.

Accent Detection & Correction:

- Helps identify speaker’s native phonetic patterns.
- Example: Indian English vs American English pronunciation differences.

Language Learning Apps:

- Duolingo or Babbel gives pronunciation feedback using articulatory patterns.

Example: Voice Assistant Confusion

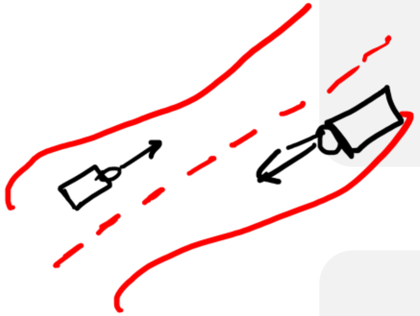
- Command: 'Turn off the light' vs. 'Turn on the light'
- Mistaking /f/ for /n/ can cause misinterpretation
- Articulation analysis helps avoid such errors

Q&A Session

1. “Can a single sound have more than one manner or place of articulation?”
2. “How do different languages affect these articulation patterns?”
3. “Why do speech systems struggle with Indian accents?”

thisclassisanaturallanguageprocessingclass

1 2 3 4 5 6 7 8



Word Boundary Detection

start & end } word

- Spaces make reading easy… but not all languages use them
- Even in English, boundaries aren't always obvious
- Core to many NLP tasks: search, translation, chatbots

Word Boundary Detection

Sentence-1

- ***This is a very easy example.***

Sentence-2

- ***Thisisaveryeasyexample.***

1 2 3 4 5 6

Speech Recog.:

↓

Let's eat grandma

↓

Let's eat, grandma.

Word Boundary Detection

Search engines

Newyorktimes

↓

New York Times

- **Word Boundary Detection :** Identifying start/end of words in text or speech
- **Also called:** Word Segmentation
- **Why needed:** preprocessing for tokenization, parsing, analysis

Importance of Word Boundary Detection

- Almost every NLP task depends on knowing words: sentiment analysis, search engines, speech recognition, machine translation.
- Without correct boundaries, downstream tasks fail.
- Example: In speech recognition, “Let’s eat, grandma” vs. “Let’s eat grandma” [punctuation & boundaries change meaning]

Word Boundary Detection

spaces → explicit boundries

→ thebigdogran

the | big | dog | ran ←

the | big | do | gran ← ✓

Languages like English have spaces to mark word boundaries, but spoken language and many Indian languages do not have clear word delimiters.

Word boundary detection identifies the beginning and end of words in continuous text or speech.

Techniques:

- Acoustic models to detect pauses in speech → Ms | ms
- Statistical models to predict likely word combinations

Applications: Speech-to-text systems, OCR, text segmentation

Example: In speech, 'I scream' and 'ice cream' may sound similar, and boundary detection resolves ambiguity.

Word Boundary Detection

bank → 2 meanings

Identifying the beginning and end of words in a continuous stream of speech or text is a crucial task in NLP and speech processing.

Challenges:

- **Continuous Speech**: In natural speech, there are often no clear pauses between words, making it hard to distinguish word boundaries.
- **Homophones**: Words that sound the same but have different meanings and spellings (e.g., "night" and "knight") add complexity.
- **Variability**: Different speakers, accents, and speaking rates can alter the way word boundaries appear.

Where it’s easy vs. hard

Easy in: English text with spaces → spaces mark boundaries.

Hard in:

- Chinese, Japanese, Thai → no spaces between words.
- Speech → no clear pauses between words.
- OCR output → might have missing or extra spaces.

Examples of Ambiguity

English example

- I scream vs. Ice cream (same sounds, different segmentation).
- Speech system must decide based on context.

Chinese Example

- "我喜欢吃苹果" → literally “I like eat apple” → no spaces; we must segment as 我 | 喜欢 | 吃 | 苹果.

Examples of Ambiguity

Hashtags & social media

- #nowthatchersdead -> Some read as “Now Thatcher’s dead” vs. “Now that Cher’s dead”.

Makeup Tricky Case

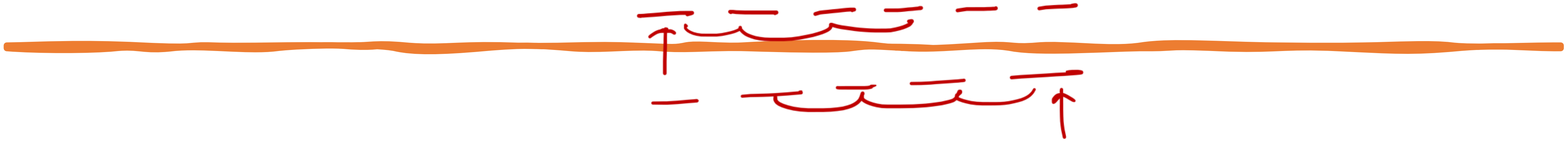
- "therapistfinder“ -> Could be therapist finder or the rapist finder.

Word Boundary Detection

Techniques:

- **Acoustic Cues**: Utilizing changes in pitch, duration, and intensity of the speech signal to infer word boundaries.
- **Phonotactic Rules**: Language-specific rules about which phoneme sequences are permissible can help identify boundaries.
- **Statistical Methods**: Probabilistic models like Hidden Markov Models (HMMs) or neural networks can predict word boundaries based on learned patterns.
- **Lexical Information**: Using dictionaries and known word lists to match spoken segments to possible words.

Approaches for Text Word Boundary Detection



Rule-based methods

- Use spaces, punctuation, capitalization, and dictionary lookups.
- Good for languages with spaces; fail in unsegmented text.

Dictionary-based maximum matching

- **Forward Maximum Matching (FMM):** Try to match the longest word from the start.
- **Backward Maximum Matching (BMM):** Start from the end and match backwards.
- **Problem:** ambiguous segmentations.
- **Example in Chinese:** "研究生命" could be “研究 | 生命” (research life) or “研究生 | 命” (graduate student’s life).

Word Boundary Detection in Speech

Challenge:	No guaranteed silence between words.
	Human speech is continuous: "I’mgoingtothemarket" → no clear gaps.
Acoustic cues:	Slight pauses, intonation changes.
	Coarticulation: sounds blend at boundaries.
Approaches:	Hidden Markov Models (HMMs) trained on phoneme sequences.
	Neural networks trained on raw audio (CTC loss for alignment).
	Example: "She sells sea shells" → model must segment and match to dictionary words.

Challenges in Real-world Data

01

Noisy Input: OCR errors, speech misrecognition.

02

Multiple valid segmentations: "unlockable" → un-lockable vs. unlock-able.

03

Code-switching: Mixing languages in the same sentence.

04

Domain-specific vocabulary: Medical, legal, gaming slang.

Modern Deep Learning Approaches

BiLSTM-CRF models: Learn to predict “B” (begin), “I” (inside), “E” (end), “S” (single) labels for characters.

Transformers (BERT-based): Context-aware segmentation.

Example:

- Input: "machinelearningisfun"
- Model predicts: machine | learning | is | fun based on context embeddings.

Advantage:

- Handles ambiguity better than rules.

Applications

MACHINE TRANSLATION → NEEDS CORRECT WORD UNITS.

SEARCH ENGINES → BETTER INDEXING & KEYWORD MATCHING.

TEXT-TO-SPEECH → NATURAL PAUSES AND INTONATION.

CHATBOTS → INTENT DETECTION DEPENDS ON WORD SEGMENTATION.

Challenge

"ilovedataandmachinelearningalot"

1 2 3 4 5 6 7 8

Attendance

20/Aug/2025

NLP → 10-11 AM

Sec-A

Sec B

Present :→ (1-20) → 013 (Sahil)

(20-40) → 228,

(40-60) →

(60-80) → 70, 66, 78,

(80-100) →

→ (234), (257), (221), (246), (1601104220097), 274,

105,

Argmax Based Computations

$\{2, 4, 10, 7, 5, 8\}$
$0, 1, 2, 3, 4, 5$
2

$\{0.4$ class0, 0.8 class1, 0.9 class2, 0.3 class3$\}$
class1

Definition: The argmax function returns the argument (input) that maximizes a given function. It's widely used in NLP for making decisions based on probabilistic models.

- **Example**:
  - **Formula**: $\text{argmax}_x P(x|y)$
    - **Explanation**: For a given input y, find the value x that maximizes the probability $P(x|y)$.
- **Steps**:
  - **Define the Probability Function**: Based on a model or data, such as a probabilistic grammar or a trained neural network.
  - **Evaluate All Possible Inputs**: Calculate the probability for each possible x.
  - **Select the Maximum**: Choose the x with the highest probability.

Sentence : → "This movie was amazing"

→ $\boxed{\text{Positive}}$ → 0.8

→ Negative → 0.1

→ Neutral → 0.2

$$\text{argmax}\left\{ \underbrace{\underline{0.8}, 0.1, 0.2}_{\text{Raw Probabilities}} \right\} \Rightarrow \underset{\text{Result}}{\underset{\downarrow}{\text{Positive}}}$$

Argmax Based Computations

Definition: The argmax function returns the argument (input) that maximizes a given function. It's widely used in NLP for making decisions based on probabilistic models.

Steps:

- **Define the Probability Function**: Based on a model or data, such as a probabilistic grammar or a trained neural network.
- **Evaluate All Possible Inputs**: Calculate the probability for each possible x.
- **Select the Maximum**: Choose the x with the highest probability.

Why Argmax in NLP?

Many NLP tasks involve probabilities.

- In text classification, we want to predict the category with the highest probability.
- In POS tagging, for each word, we want the tag that maximizes the probability given the context.
- In Machine Translation, at each step, we want the word with the highest likelihood according to our language model.

Basically, argmax is the way a model says: ‘Out of all possible answers, I choose this one, because it’s most likely to be correct.’

Example -1: Text Classification

Suppose we are building a classifier to detect the sentiment of a sentence:

Sentence: '*This movie was amazing!*'

Now our model gives the following probabilities:

- Positive: 0.8
- Negative: 0.1
- Neutral: 0.1

If I ask:

$$\text{argmax}_{label}\{0.8, 0.1, 0.1\}$$

The result is **Positive**.

So the model's prediction is 'Positive sentiment'.

Notice how argmax converted **raw probabilities** into a **decision**."

Example 2: Part-of-Speech Tagging

Suppose we have a sentence: '*Dogs bark loudly*'.

For the word 'bark', the model may consider multiple tags:

- Verb (probability = 0.7)
- Noun (probability = 0.3)

If we use argmax here:

$$\text{argmax}_{tag}\{0.7, 0.3\} = \text{Verb}$$

So the system predicts 'bark' as a **Verb**, because it has the maximum probability.

This is how modern NLP systems disambiguate words."

Hidden Markov Models (HMM) and Speech Recognition

HMM and Speech Recognition

Definition: Hidden Markov Models (HMMs) are statistical models used to represent time-series data like speech. Each state in an HMM represents a phoneme or word and transitions between states are governed by probabilities.

Components:

- **States**: Represent different parts of the speech signal, such as different phonemes or words.
- **Observations**: The actual speech sounds (acoustic signals) that are heard.
- **Transition Probabilities**: The probabilities of moving from one state to another.
- **Emission Probabilities**: The probabilities of producing a particular observation from a state.

HMM and Speech Recognition…

Speech Recognition:

- **Definition**: The process of converting spoken language into text.
- **Process**:
  - **Feature Extraction**: Convert the speech signal into a sequence of feature vectors that represent the important characteristics of the sound.
  - **Modeling**: Use HMM to model the sequence of feature vectors as a sequence of states.
  - **Decoding**: Find the most likely sequence of words given the observed feature vectors using algorithms like the Viterbi algorithm.

HMM and Speech Recognition…

Example:

- **Training**: Train the HMM on a labeled dataset of speech and corresponding text. This involves estimating the transition and emission probabilities from the data.
- **Recognition**: Use the trained HMM to recognize new speech input. Given a new sequence of feature vectors, the HMM finds the most probable sequence of states (words) that could have generated the observed sequence.

HMM and Speech Recognition…

Use Case in NLP:

- Part-of-Speech Tagging
- Named Entity Recognition
- Speech Recognition
- Example: Given a sequence of words, we use argmax to choose the most probable tag sequence.

What is an HMM?

Hidden States: phonemes (/h/, /e/, /l/, /o/)

Observations: acoustic features (MFCCs, frequency patterns)

Transition Probabilities: likelihood of moving between phonemes

Emission Probabilities: likelihood of observing a signal given a phoneme

Initial Probabilities: likelihood of starting in a phoneme

Analogy and Example

Analogy:

Curtain example: you can’t see actions (hidden states)

But you hear sounds (observations)

Goal: infer hidden states from observations

Example:

Hidden States: Sunny, Rainy

Observation: Umbrella (yes/no)

Sunny → Umbrella (0.1), Rainy → Umbrella (0.8)

HMM infers most likely weather sequence from umbrella data

Speech Recognition with HMM

Step 1: Convert audio into frames (MFCC features)

Step 2: Each phoneme modeled as an HMM

Step 3: Use Viterbi algorithm to align observed features to phonemes

Step 4: Map phoneme sequence to words

Real-World Applications

- Early Google Voice Search
- IBM Watson (Jeopardy system)
- Automatic transcription in call centers
- Voice commands in cars & GPS systems